Answers learning task 14

2026 – 2027

Exercise 8.29: A postmix beverage machine is adjusted to release a certain amount of syrup into a chamber where it is mixed with carbonated water. A random sample of 25 beverages was found to have a mean syrup content of $\bar{x} = 30$ mL and a standard deviation of $s = 0.5$ mL. Find a $95\,\%$ CI on the mean volume of syrup dispensed.

Answer

$[29.7936, 30.2064]$

Exercise 8.37: The compressive strength of concrete is being tested by a civil engineer who tests 10 specimens and obtains the following data:

2216	2237	2249	2204
2225	2301	2281	2263
2318	2255		

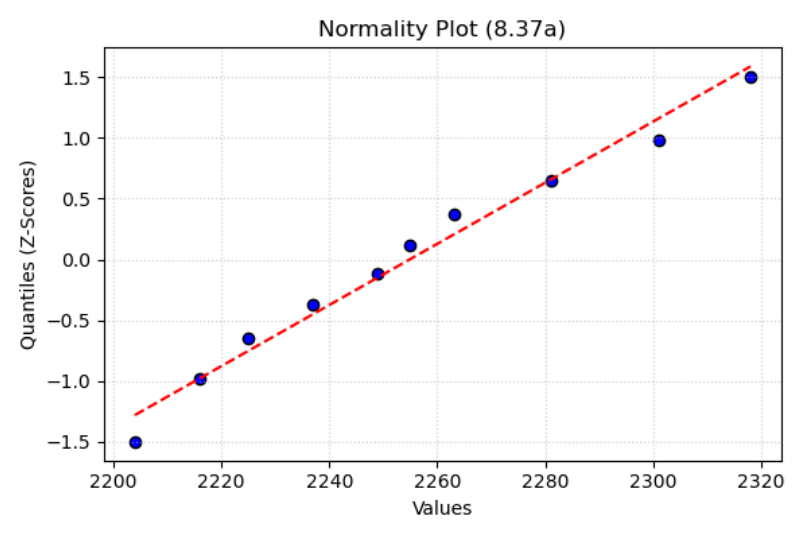
(a) Check the assumption that compressive strength is normally distributed. Include a graphical display in your answer.

(b) Construct a $95\,\%$ two-sided confidence interval on the mean strength.

(c) Construct a $95\,\%$ lower confidence bound on the mean strength. Compare this bound with the lower bound of the two-sided confidence interval and discuss why they are different.

Answer

(a) Normal probability plot:



The points lie reasonably close to a straight line, indicating that the compressive strength data can be assumed to be normally distributed.

(b) $[2228.5546, 2281.2454]$

(c) $[2233.5513, \infty)$

Exercise 8.39: An article in *Computers & Electrical Engineering* ["Parallel Simulation of Cellular Neural Networks" (1996, Vol. 22, pp. 61–84)] considered the speedup of cellular neural networks (CNN) for a parallel general-purpose computing architecture based on six transputers in different areas. The data follow:

3.775302	3.350679	4.217981	4.030324	4.639692
4.139665	4.395575	4.824257	4.268119	4.584193
4.930027	4.315973	4.600101		

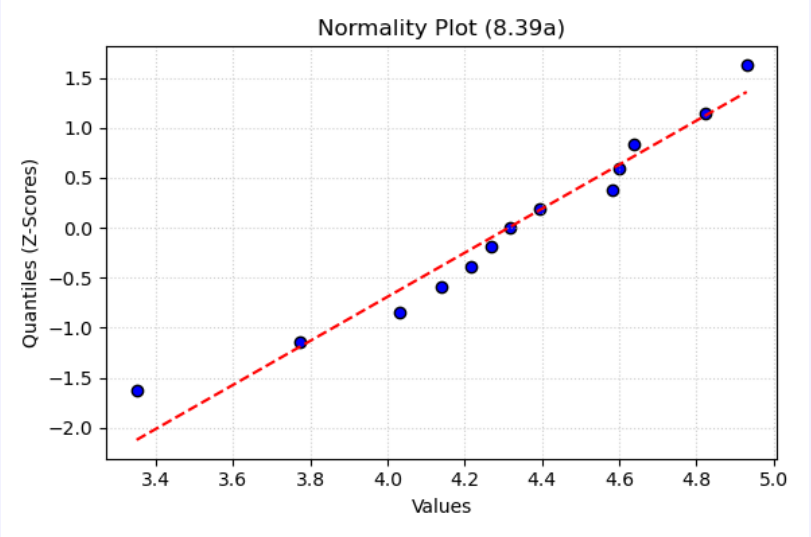
(a) Is there evidence to support the assumption that speedup of CNN is normally distributed? Include a graphical display in your answer.

(b) Construct a $95\,\%$ two-sided confidence interval on the mean speedup.

(c) Construct a $95\,\%$ lower confidence bound on the mean speedup.

Answer

(a) Normal probability plot:



The points approximate a straight line, showing that the speedup is normally distributed.

(b) $[4.0517, 4.5748]$

(c) $[4.0993, \infty)$

Exercise 8.41: An article in *Nuclear Engineering International* (February 1988, p. 33) describes several characteristics of fuel rods used in a reactor owned by an electric utility in Norway. Measurements on the percentage of enrichment of 12 rods were reported as follows:

2.94	3.00	2.90	2.75	3.00	2.95
2.90	2.75	2.95	2.82	2.81	3.05

(a) Use a normal probability plot to check the normality assumption.

(b) Find a $99\,\%$ two-sided confidence interval on the mean percentage of enrichment. Are you comfortable with the statement that the mean percentage of enrichment is 2.95%? Why?

Answer

(a) The data appear to be normally distributed.

(b) $[2.8126, 2.9907]$, yes, because $2.95 \in$ CI.